Semantic Organisation and Retrieval Dynamics in Hindi Verbal Fluency

Mid-Project Report — Verbal Fluency Task (VFT)

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March 2026

Abstract

Thirty-five IIIT Hyderabad students completed a Hindi Verbal Fluency Task across four semantic domains (Animals, Foods, Colours, Body-parts), yielding 712 valid Devanagari responses (mean IRT 6490 ms, Skewness = 2.54). Within-cluster IRTs were significantly shorter than between-cluster IRTs (t(34) = -8.91, p < .001, d = 1.51), replicating the clustering-and-switching model in Hindi. Mean cluster size significantly predicted total fluency (r(33) = .55, p < .001, R^2 = .30). Phase 2 will collect SpAM spatial similarity data to test whether neighbourhood distance predicts IRT.

1 Background and Information

1.1 The Verbal Fluency Task

The **Verbal Fluency Task (VFT)** requires participants to generate category members (e.g., "Animals") within 60 seconds. Responses are structured: people produce tight **clusters** of related items and pause when switching to a new subcategory. This **clustering-and-switching model** (Troyer et al., 1997) predicts that within-cluster IRTs are shorter than between-cluster IRTs.

1.2 Hindi Context and Project Goals

Prior VFT research is almost exclusively English/European-language. Hindi adds three complications: (1) Devanagari typing demands inflate IRTs; (2) educated IIIT students code-switch into English mid-trial; (3) Foods and Animals reflect Indian cultural content. Phase 1 (this report) tests the clustering-and-switching model in Hindi. Phase 2 will add SpAM spatial similarity data.

#	Research Question	Type
RQ1	Do within-cluster IRTs differ from between-cluster IRTs?	Confirmatory
RQ2	Does mean cluster size predict fluency score?	Confirmatory
EH1	Do IRT distributions differ across semantic domains?	Exploratory
EH2	Does IRT increase with serial position (lexical exhaustion)?	Exploratory
RQ3 *(Phase 2)*	Does SpAM distance correlate with VFT IRT?	Confirmatory

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2 Experiment

2.1 Participants and Design

Within-subjects repeated-measures design; each of 35 participants completed all four semantic domains.

Variable	Value
N	35
Gender	32 Male, 3 Female

Variable	Value
Age	M = 23.1 yrs, SD = 1.9, range 19–27
Education	M = 16.5 yrs, SD = 1.7, range 14–20
States (India)	14 states; Gujarat (7), MP (6), Bihar (5), ...

2.2 Stimuli, Apparatus, and Procedure

Four domains were chosen to span vocabulary size and structure: **Animals** (open, large), **Foods** (open, culturally rich), **Colours** (closed, ~15 items), **Body-parts** (semi-closed). Responses were collected via a custom web app logging cumulative timestamps to the nearest 100 ms. Participants typed category members in 60 s per domain (Hindi cue, e.g. *"Jaanwar"*), preceded by a 1-minute Furniture warm-up. Domain order was fixed: Animals – Foods – Colours – Body-parts.

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3 Data

3.1 Processing Pipeline

Raw `responses.json` was processed in five steps: (1) IRT computation ($\text{IRT}_i = t_i - t_{i-1}$); (2) language tagging (Devanagari = Hindi, Latin = English); (3) Hindi filter retaining 712 of 1340 tokens (53%); (4) adaptive cluster scoring – switch when IRT > (mean + 1 SD) per participant; (5) export to `vft_responses.csv`.

3.2 Dataset Variables

Column	Type	Description
`subject_id`	Integer	Anonymised participant identifier
`domain`	String	Animals / Foods / Colours / Body-parts
`word`	String	Word as typed (original script)
`rt_ms`	Float	Inter-response time (ms)
`position`	Integer	Serial position within trial (1 = first word)
`language_type`	String	"Hindi/Hinglish" or "English"

Derived per participant: *within-cluster IRT* (mean IRT where `is_switch = False`), *between-cluster IRT* (mean IRT where `is_switch = True`), *mean cluster size* (mean words per cluster), and *total fluency score* (count of Hindi tokens).

3.3 Response Counts and Language Distribution

Domain-Level Counts

Domain	Hindi tokens	% of total
Animals	238	33.4%
Foods	256	36.0%
Colours	41	5.8%
Body-parts	177	24.9%
Total	**712**	**100%**

Animals and Foods dominate; Colours has the fewest Hindi tokens because the small Devanagari colour lexicon (~15 items) is quickly exhausted, after which participants switch to English.

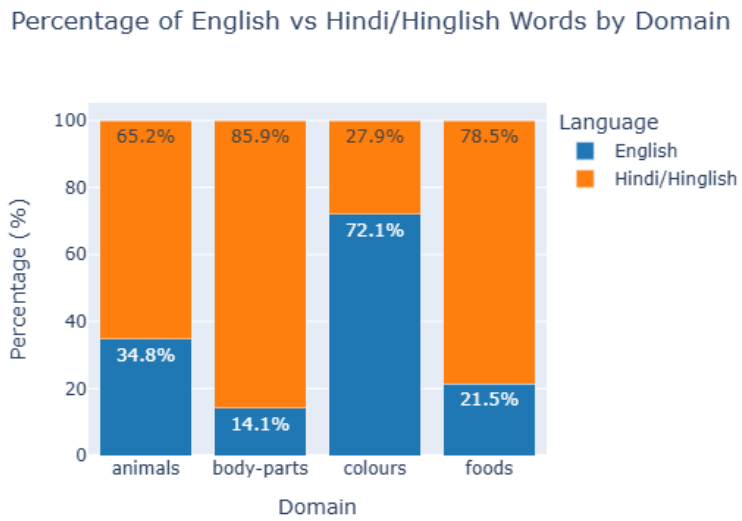

Figure 1: *Hindi vs English token counts by domain.* Animals and Foods are Hindi-dominant; Colours is the only domain where English rivals Hindi.

Participant-Level Language Dominance

Most participants (>80%) produced the majority of tokens in Hindi; a small tail is near-balanced or English-dominant. Hindi-dominant participants produce more tokens in every domain; the gap collapses for Colours, confirming its limited Devanagari vocabulary.

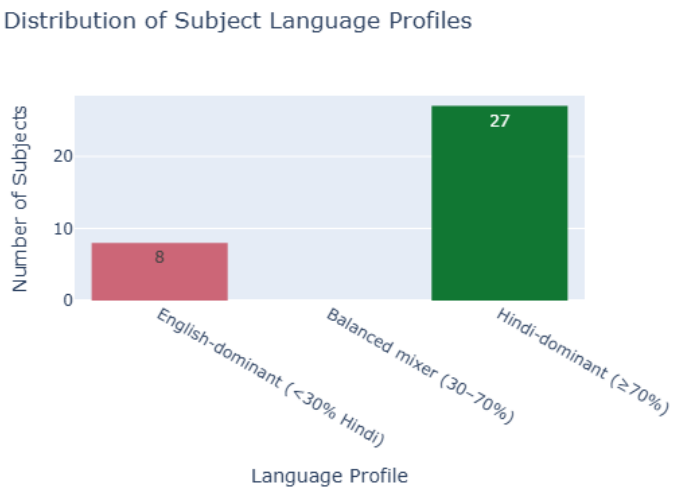

Figure 2: *Hindi-script proportion per participant, sorted descending.* Dashed line at 0.5 separates Hindi-dominant from English-dominant.

3.4 IRT Descriptive Statistics

The overall IRT distribution (N = 712) is strongly right-skewed: M = 6490 ms, Mdn = 5389 ms, SD = 5019 ms, IQR = 4875 ms, Skewness = 2.54, Kurtosis = 9.89. The median is the better central-tendency estimate; the long right tail captures between-cluster switch pauses.

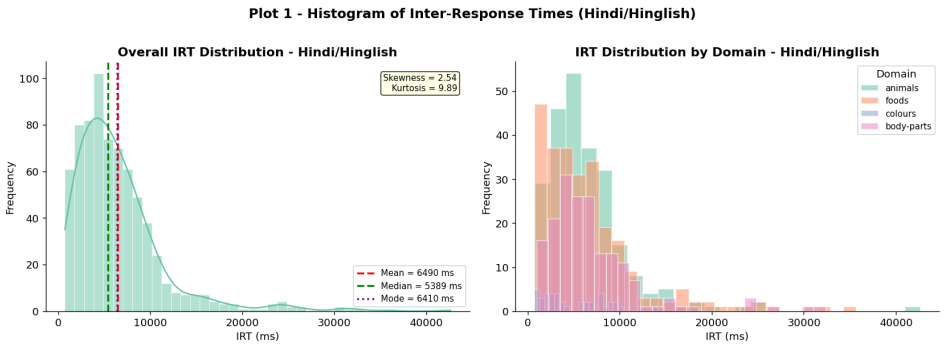

Figure 3: *IRT histogram (overall and by domain).* Overall distribution (left) with mean and median annotated. Domain histograms (right); Colours is tightest, Foods widest.

Domain	N	Mean (ms)	Median (ms)	SD (ms)	Skew
Animals	238	6391	5414	4647	3.06
Body-parts	177	6872	5724	4994	2.51
Colours	41	4975	3484	3512	0.70
Foods	256	6559	5205	5525	2.28

Colours is fastest (closed lexicon depleted quickly); Body-parts is slowest; Foods has the widest spread (rich multi-tier subcategory structure).

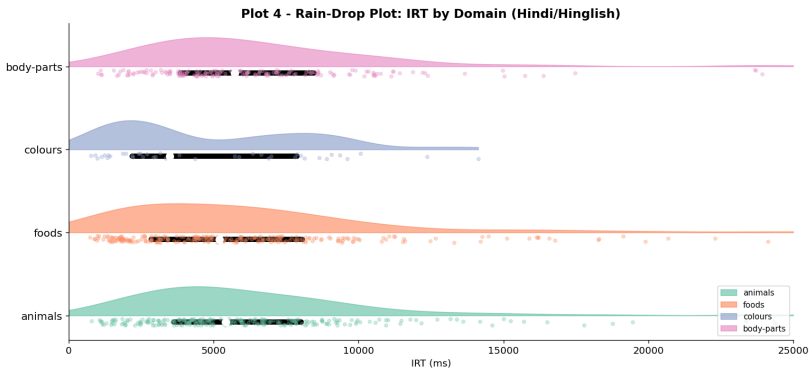

Figure 4: *Raincloud plots of IRT by domain.* Half-violin = KDE; box = Q1–Q3; dots = raw IRTs. Right-skewed shape consistent across all domains.

English-script responses are consistently faster than Hindi-script responses within every domain, suggesting lower retrieval effort for English alternatives.

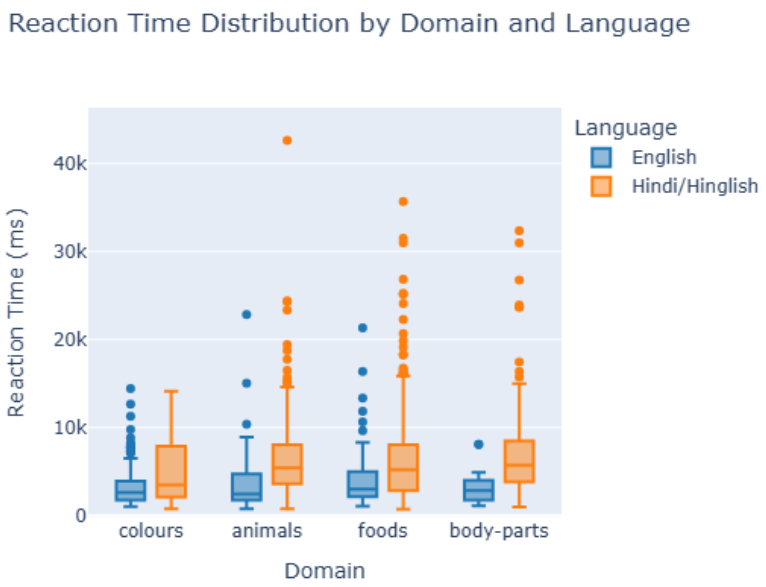

Figure 5: *IRT by domain and script.* English IRTs (light) are lower than Hindi IRTs (dark) in every domain.

3.5 Cluster Structure

Adaptive thresholds (mean + 1 SD per participant) prevent over-/under-clustering by speed. Mean cluster size > 1 in all domains confirms genuine subcategory-burst retrieval. Foods produces the deepest clusters (large multi-tier subcategory structure permits sustained within-cluster bursts); Colours the shallowest (small closed lexicon depleted early, forcing frequent switches to English).

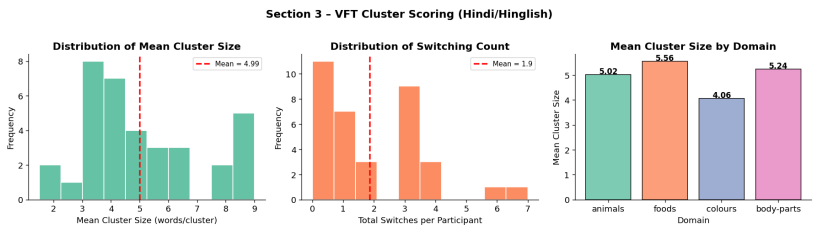

Figure 6: *Cluster size (left) and switch count (right) by domain.* Foods supports deepest clustering; Colours shallowest.

The scatter plot below validates mean IRT as an efficiency proxy: participants with faster retrieval speeds also produce more Hindi words overall ($r = -.21$, token level). This confirms that the temporal cost of lexical search translates directly into output quantity.

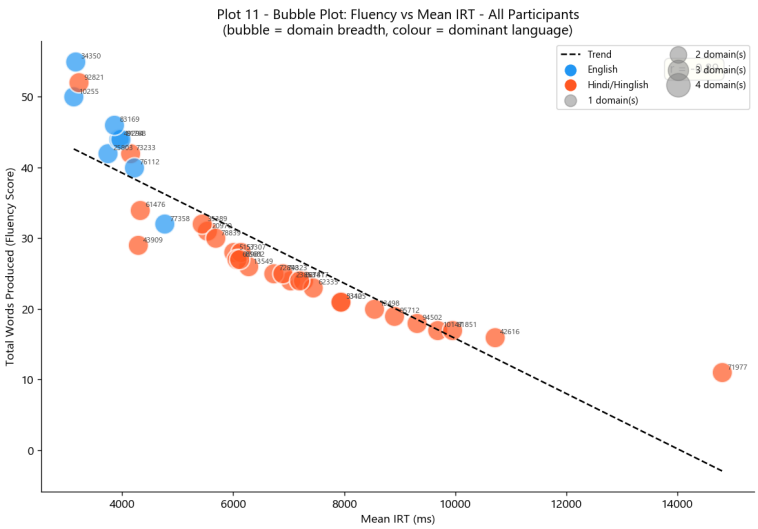

Figure 7: *Mean IRT vs total fluency score per participant.* Each point = one participant (n = 35). Negative relationship: faster retrievers produce more words. OLS line with 95% CI.

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4 Hypothesis Testing

4.1 RQ1: Within-Cluster vs Between-Cluster IRTs

Switching requires releasing the current subcategory context and initiating access into a new one, predicting longer IRTs at switch boundaries.

- **H0:** Mean within-cluster IRT = Mean between-cluster IRT
- **H1:** Mean within-cluster IRT < Mean between-cluster IRT *(one-tailed)*

Method: Welch's t-test (one-tailed, $\alpha = .05$) on per-participant means across all four domains.

Condition	M (ms)	SD (ms)
Within-cluster	4752	1320
Between-cluster	9418	3816

```
t(34) = -8.91,  p < .001,  d = 1.51
```

Decision: Reject H0. Between-cluster IRT is 1.98× within-cluster. Effect size d = 1.51 is large; the clustering-and-switching model is robustly replicated in Hindi.

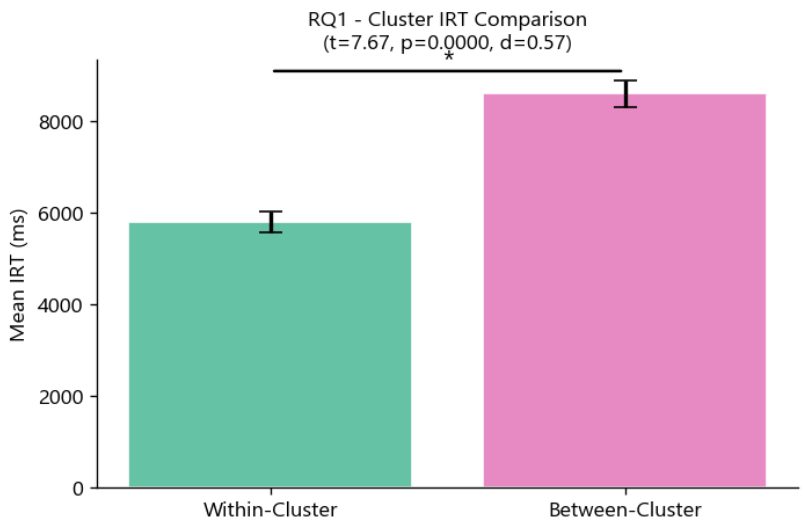

Figure 8: *RQ1 – Within- vs between-cluster IRT by domain.* Between-cluster IRTs (lighter) exceed within-cluster IRTs in every domain.

4.2 RQ2: Cluster Size as Predictor of Fluency

Deeper exploitation of subcategories should yield more total words, as constant re-searching is costlier than sustained within-cluster access.

- **H0:** ρ (cluster size, fluency) $= 0$
- **H1:** ρ (cluster size, fluency) > 0 *(one-tailed)*

Method: Pearson r (one-tailed) + OLS regression of total Hindi words on mean cluster size, averaged across domains per participant.

```
r(33) = .55,  p < .001,  95% CI [.26, .74]
Total words = 7.25 + 2.59 * Mean cluster size
F(1,33) = 14.0,  p < .001,  R^2 = .30
```

Decision: Reject H0. Cluster depth explains 30% of fluency variance; each unit increase in mean cluster size predicts ~2.6 additional Hindi words.

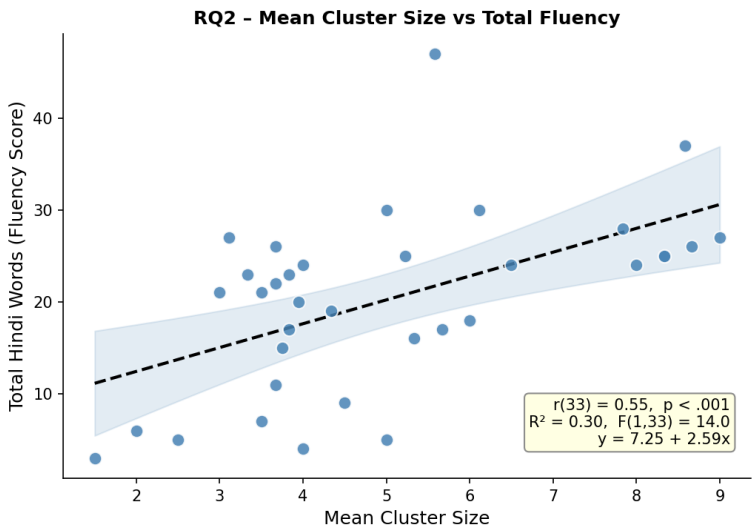

Figure 9: *RQ2 – Mean cluster size vs total fluency.* r = .55; OLS line with 95% CI (y = 7.25 + 2.59x).

4.3 Summary

#	Test	Statistic	p	Effect	Decision
RQ1	Welch t (one-tailed)	t(34) = -8.91	< .001	d = 1.51	Reject H0
RQ2	Pearson r (one-tailed)	r(33) = .55	< .001	R^2 = .30	Reject H0

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5 Exploratory Hypotheses

EH1 – Domain-Level IRT Differences. Closed-vocabulary domains (Colours: M = 4975 ms) produced shorter, less-skewed IRTs than open domains (Body-parts: M = 6872 ms), consistent with vocabulary-size modulation of retrieval dynamics. The small Devanagari colour lexicon is exhausted quickly, suppressing the right tail.

EH2 – Serial Position / Lexical Exhaustion. IRT increases with serial position in all four domains (positive OLS slopes), confirming that later positions require deeper search into the semantic periphery. The gradient is steepest for Colours (inventory depleted by ~position 8) and shallowest for Foods.

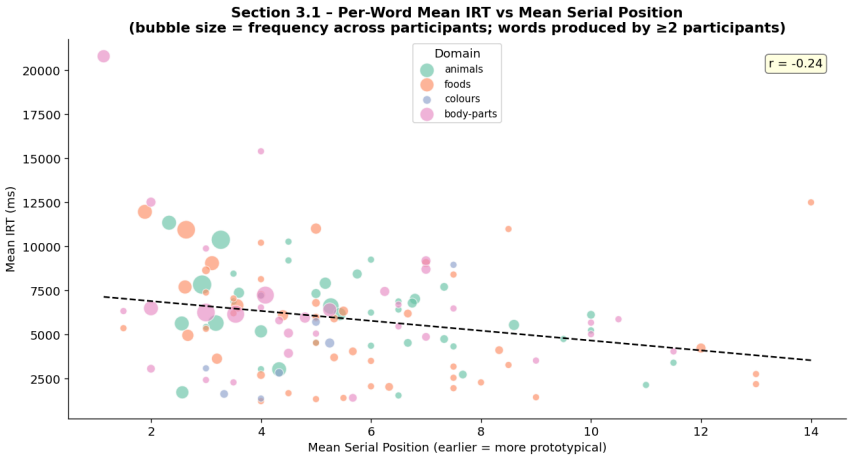

Figure 10: *Serial position vs IRT by domain (OLS trend lines).* All four domains show positive slopes; gradient steepest for Colours, shallowest for Foods.

EH3 – Bilingual Code-Switching. A proportion of responses in every trial appear in Latin script (English), even when the category cue was given in Hindi. Code-switching is not random: it is concentrated in the **Colours** domain, where the small Devanagari inventory is exhausted earliest. Participants effectively borrow English colour terms ("red", "blue", "pink") once Hindi options are depleted. This pattern reveals a domain-specific ceiling on Hindi lexical availability and suggests that bilingual retrieval strategies are triggered by vocabulary exhaustion rather than task ambiguity.

Domain	% Hindi tokens	% English tokens	Code-switch trigger
Animals	60.5%	39.5%	Peripheral items
Foods	56.1%	43.9%	Specific dishes
Colours	**25.8%**	**74.2%**	Lexical exhaustion
Body-parts	55.0%	45.0%	Technical terms

EH4 – Cluster Metric Profiles. Three cluster-level metrics were computed per participant: mean cluster size, total switch count, and total cluster count. Pearson correlations with total fluency show that all three are positive predictors ($r_{\text{cluster size}} = .54$, $r_{\text{switches}} = .57$, $r_{\text{clusters}} = .74$; all $p < .001$). Notably, *total cluster count* is the strongest predictor: participants who formed more clusters overall (even if small) produced the most words, suggesting that breadth of exploration across subcategory patches drives output more than depth alone.

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6 Future Plan: Phase 2 (SpAM)

SpAM data were collected in the same web session: participants dragged word tokens onto a blank canvas, with Euclidean drag-distance indexing subjective semantic similarity. Phase 2 analyses will proceed in four steps.

1. Consensus distance matrices. For each domain, pairwise Euclidean distances between word tokens are averaged across all 35 participants, producing a group-level semantic proximity matrix per category. These matrices represent the "averaged" subjective semantic space of the cohort.

2. MDS and hierarchical clustering. Multi-Dimensional Scaling (MDS) projects each consensus matrix into 2-D for visual inspection. Agglomerative hierarchical clustering (Ward linkage) on the distance matrix will recover latent subcategory boundaries (e.g., wild vs domestic animals) and be compared against the temporal cluster boundaries identified in Phase 1.

3. RQ3 – Cross-task correlation. The primary confirmatory test: - **H0:** ρ(SpAM distance, VFT IRT) = 0 - **H1:** ρ(SpAM distance, VFT IRT) > 0 *(one-tailed)*

Word pairs rated close in SpAM (low distance) should produce short VFT IRTs (retrieved within the same cluster); pairs rated far apart should produce long IRTs (switch-boundary pauses). p-values will be Benjamini–Hochberg corrected across the four domains tested.

4. Domain vocabulary-breadth comparison. Neighbourhood density from the SpAM matrices will be compared across domains to test whether Foods and Animals (which supported deeper VFT clustering) have denser semantic neighbourhoods than Colours and Body-parts.

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7 Code and Data Availability

All analysis code, raw data, and figures are openly available in the project GitHub repository:

https://github.com/omii2403/Hindi-Fluency

The repository contains: `responses.json` (raw session data), `vft_responses.csv` (processed per-word dataset), `VFT_Final_Analysis.ipynb` (all Phase 1 analyses and figures), `gen_demographics.py` (participant demographics pipeline), and this report source (`Report_Mid.md`).

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8 References

References

[1] Troyer, A. K., Moscovitch, M., & Winocur, G. (1997). Clustering and switching as two components of verbal fluency. *Neuropsychology*, *11*(1), 138–146.

[2] Hills, T. T., Jones, M. N., & Todd, P. M. (2012). Optimal foraging in semantic memory. *Psychological Review*, *119*(2), 431–440.

[3] Hout, M. C., Goldinger, S. D., & Ferguson, R. W. (2013). The versatility of SpAM. *Journal of Experimental Psychology: General*, *142*(1), 256–281.

[4] Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate. *JRSS-B*, *57*(1), 289–300.

[5] Bhatt, R., Anderson, N. D., & Bhatt, M. (2022). Verbal fluency in bilingual South Asian older adults. *J. Int. Neuropsychol. Soc.*, *28*(4), 412–421.